Example 11

Three-Story Plane Frame with Viscous Damper Elements - Nonlinear Time History Analysis

Problem Description

The El Centro 1940 (N-S) record is used in the nonlinear time history analysis. Three viscous damper elements of the type described in Hanson (1993) are used to connect the chevron braces to the frame. Two models are investigated. In the first model, the damper elements are intended to produce about 5% damping in the fundamental mode. In the second model, damping is increased to 25%.

The ETABS viscous damper element (DAMPER) is a uniaxial damping device with a linear or nonlinear force-velocity relationship given by $F = CV^{\alpha}$.

The damper elements are modeled in ETABS by assigning a panel zone with a nonlinear link property to the mid-span point object where the chevrons intersect the beams at each story. The link properties use the uniaxial damper property (DAMPER) and provide beam-brace connectivity with nonlinear behavior in the U2 (shear in the 1-2 plane) direction. Under this arrangement, displacements are transferred between the chevrons and the frame via the link elements (dampers) undergoing shear deformation.

The time increment for output sampling is specified as 0.02 second.

Geometry, Properties and Loads

This is a single-bay, three-story plane frame subjected to ground motion, as shown in Figure 11-1. The frame is modeled as a two-column line, one-bay system. Kip-inch-second units are used. The modulus of elasticity is taken as 29000 ksi. Column, beam and brace section properties are user defined.

A single rigid diaphragm is allocated to each story level and connects all three point objects (two column points and one mid-span point) at each story. Because of the rigid diaphragms, no axial force will occur in the beam members. All members are assigned a rigid zone factor of 1.

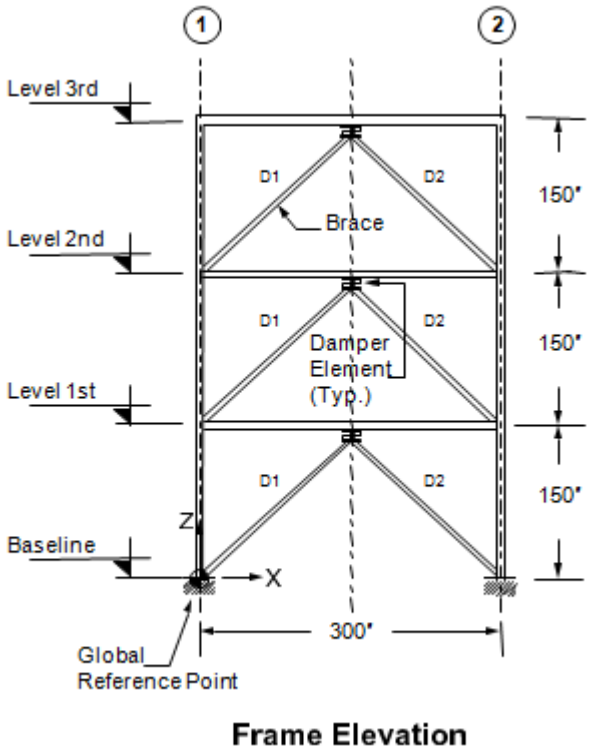

Figure 11-1 Planar Frame with Damper Elements

Technical Features of ETABS Tested

- Two-dimensional frame analysis
- Use of panel zones
- Use of uniaxial damper elements
- Point assignments
- Nonlinear time history analysis
- Ritz vectors

Results Comparison

Sample results for $\alpha = 1$ are compared in Table 11-1 with results from the nonlinear analysis program DRAIN-2DX (Prakash, et al. 1993) for both 5% and 25% damping cases.

Table 11-1 Results Comparison

Level	5% Damping		25% Damping	
	ETABS	DRAIN-2DX	ETABS	DRAIN-2DX
Comparison of Maximum Story Deflections				
3rd	4.09	4.11	2.26	2.24
2nd	3.13	3.14	1.75	1.71
1st	1.63	1.63	0.89	0.87
Comparison of Maximum Link Shear Force				
3rd	6.16	5.98	14.75	14.75
2nd	10.79	10.80	32.82	32.84
1st	15.15	15.02	44.90	44.97
Comparison of Maximum Brace Axial Force				
3rd	4.36	4.23	10.43	10.43
2nd	7.63	7.63	23.21	23.22
1st	10.71	10.62	31.75	31.80

Computer File

The input data files for this example are Example 11A.EDB (5% damping) and Example 11B.EDB (25% damping). The time history file is ELCN-THE. These files are provided as part of the ETABS installation.

Conclusions

The comparison of results shows acceptable agreement between ETABS and DRAIN-2DX.